

WASP-104b

R-2510

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_180319 · created 2026-08-25T00:14:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458196.6282 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.043
Transit depth (Rp/Rs)^2	2.0 +/- 1.23 [%]
Orbital Inclination (inc)	84.26 +/- 2.75
Airmass coefficient 1 (a1)	0.974 +/- 0.013
Airmass coefficient 2 (a2)	0.022 +/- 0.021
Scatter in the residuals of the lightcurve fit is	1.37 %
Best Comparison Star	#3 - [165, 406]
Optimal Aperture	2.91
Optimal Annulus	10.84
Transit Duration (day)	0.047 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.043 vs 0.1214 (deviation 0.48σ)
Duration fit vs published	0.047 d vs 0.0742 d (deviation 1.81σ)
Mid-time O–C	3.6 min
Depth SNR	3.3
Residual scatter	1.37 %

Light curve

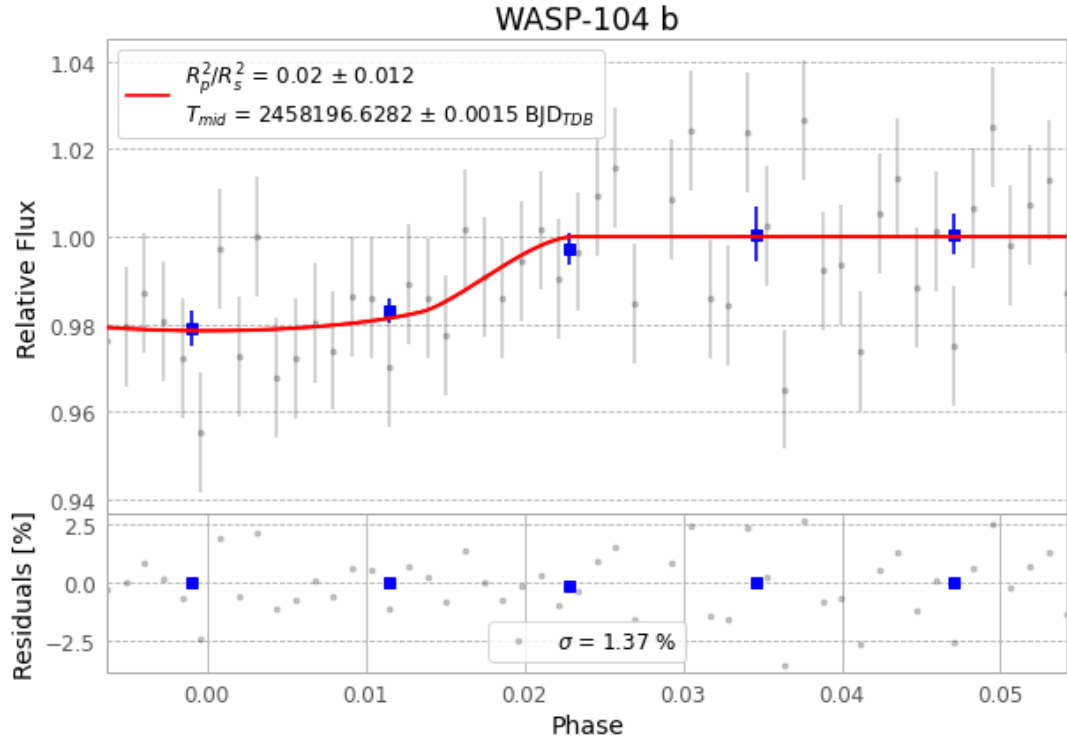


Figure 1 — FinalLightCurve_WASP-104 b_2018-03-18.png (SHA-256 30777d29b0720cf6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 190], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6ce313a02ebf6f13...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

51 FITS frames (33 MB), WASP-104180319023912.FITS → WASP-104180319051212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180319.log (8c2e29b79f59...), FinalLightCurve_WASP-104 b_2018-03-18.png (30777d29b072...), FinalLightCurve_WASP-104 b_2018-03-18.csv (243aaece4f63...)

Compute resources

Wall time 138.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:14Z.